

Deep Reinforcement Learning Under Partial Observability: Co-Simulation and Penetration-Aware Policy Learning for Traffic Signal Control

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Abstract—The integration of Connected Vehicles (CVs) and Deep Reinforcement Learning (DRL) holds immense potential for optimizing Traffic Signal Control (TSC). However, existing DRL-based TSC methods assume full connectivity and simplified communication models, limiting their real-world applicability. Partial CV penetration and unreliable V2X communication introduce severe partial observability, the impact of which remains insufficiently understood due to a lack of integrated simulation environments. To address this, we introduce an open-source co-simulation framework integrating LibSignal, SUMO, OMNeT++, Veins, INET, and Simu5G for synchronized traffic and C-V2I communication modeling. Furthermore, we propose a penetration-rate-aware adaptation mechanism that dynamically selects penetration-specific pre-trained DRL policies based on real-time estimates of CV adoption, ensuring robust control under varying observability conditions. While our framework is general and supports any learning-based TSC, we evaluate it using PressLight and MPLight, on a 4×4 urban grid. Results show that low CV penetration severely degrades baseline performance; however, our adaptation mechanism significantly increases network throughput and reduces average travel time in low-connectivity regimes. Both models benefit substantially from adaptive policy selection. Furthermore, experiments demonstrate a strong coupling between V2I link reliability and TSC effectiveness. This work highlights the critical need to model communication constraints in DRL-based TSC, offering a practical bridge toward fully connected systems. Our code and pre-trained models are available at: <https://github.com/pebeigi/PAPL-TSC>.

Index Terms—Deep Reinforcement Learning, Traffic Signal Control, OMNeT++, SUMO

vary over time [1], thereby requiring adaptive models capable of responding to evolving conditions. Advancements in intelligent transportation systems have increasingly focused on adaptive Traffic Signal Control (TSC) as a key mechanism for reducing congestion, improving travel times, and enhancing urban mobility. In recent years, Deep Reinforcement Learning (DRL) has emerged as a powerful paradigm for network-level TSC, enabling adaptive and scalable control strategies that outperform traditional rule-based methods in complex urban settings. Frameworks such as PressLight [2], CoLight [3], and others have demonstrated significant promise in both isolated intersections and large-scale grid environments.

Despite this progress, two major limitations persist. First, most DRL-based TSC models assume full vehicle-to-infrastructure connectivity. Because real-world deployment of Connected Vehicles (CVs) remains limited and uneven, the robustness of these models under partial connectivity is poorly understood. Second, although DRL-based TSC inherently relies on vehicle-to-infrastructure (V2I) communication, the volume and characteristics of the resulting network traffic remain poorly understood. In particular, the implications of communication delay, packet loss, and bandwidth constraints on control stability and overall system performance have received limited attention. It is worth noting here that the reliability of V2I communication can be compromised by network instability or distortions [4], [5]. One reason for this gap is the lack of integrated simulation platforms that jointly capture traffic dynamics, communication behavior, and learning-based control.

I. INTRODUCTION

Network-level transportation problems are often confounded by latent traffic patterns and complex network dynamics that

To address these issues, we propose a co-simulation platform that integrates LibSignal [6], an open-source DRL-based TSC framework, with the OMNeT++ [7] simulation ecosystem. The platform leverages Veins [8] for OMNeT++–SUMO [9] co-simulation, INET framework [7] for the Internet protocol stack modeling, and Simu5G [10] for cellular network simulation. This work supports simultaneous simulation of vehicle movement, control decisions, and V2I communication protocol stack across varying levels of connectivity. The proposed framework is model-agnostic with respect to the DRL-based TSC controller: PressLight and MPLight are used as representative examples to evaluate whether penetration-aware adaptation improves robustness under partial observability on the Hangzhou 4×4 traffic grid [6]. Our setup incorporates realistic assumptions about roadside unit (RSU) placement and hardware constraints, enabling a detailed analysis of how partial observability and communication overhead influence control effectiveness.

The contributions of this paper are: (1) the introduction of a scalable and extensible co-simulation framework that bridges traffic control and network communication in DRL-based TSC inference; (2) a lightweight adaptation mechanism that enhances existing DRL models by dynamically adjusting their neural network parameters based on real-time Penetration Rate (PR) estimates. Our findings highlight the importance of accounting for communication dynamics when designing and deploying learning-based TSC systems, and open the door to future research on hybrid learning-communication optimization, and infrastructure planning.

II. LITERATURE REVIEW

Early research on DRL-based network-level TSC primarily assumed full observability of the traffic state, typically via complete detector coverage or simulation-level knowledge. Busch et al. [11] showed that DRL agents with V2I knowledge significantly outperform traffic-state-agnostic agents on a 3×3 grid; however, they assumed perfect communication reliability without modeling V2I transmissions. Models such as IntelliLight [12], PressLight [2], and CoLight [3] optimize signal phases using full vehicle queue lengths or traffic flows at each intersection. More recent approaches like MPLight [13] and DACT [14] further improve scalability under full observability through distributed coordination or attention mechanisms. Vieira et al. [15] integrated V2I communication and DRL on two adjacent intersections, but their reliance on Visible Light Communication (VLC) requires dense infrastructure and limits scalability compared to Cellular-V2X. While these methods demonstrate strong performance, they implicitly rely on ideal sensing assumptions that do not reflect real-world deployments. To overcome this, recent research explores DRL under partial vehicle connectivity [5], where performance fluctuates with varying PRs, emphasizing the need for robust traffic state reconstruction from sparse observations [16].

Zhang et al. [17] presented an early DRL model functioning under partial observability, achieving reduced waiting times using only a subset of detected vehicles. Similarly,

several frameworks demonstrate strong delay reductions at low CV penetrations (10%–20%), such as the detector-free framework by Feng et al. [18], the adaptive trajectory-based scheme by Wang et al. [19], and the reduced-state Deep Q-Network (DQN) by Ducrocq and Farhi [20]. Further isolated-intersection studies include a transfer-based DQN for mixed traffic [21] and a two-stage DRL model addressing pedestrian-vehicle conflicts [22]. While promising, these models remain confined to single intersections and often rely on simplifying assumptions (e.g., Poisson arrivals [18] or static cooperation [22]). Moving beyond isolated intersections, Islam et al. [23] combined CV and traditional detector data to estimate full traffic states across four sequential intersections; however, the approach relied on traditional optimization rather than scalable DRL coordination.

To address partial observability in complex urban contexts, Zhao et al. [24] proposed CVLight for a 5×5 intersection network. Although inference uses only CV data, the critic is trained with full vehicle information, limiting the realism of assumed deployment conditions. Guo et al. [25] introduced CoTV, a scalable multi-agent framework controlling both signals and leading CVs with robust performance at 10–30% penetration; however, the model assumes ideal V2X communication. A hybrid learning-based approach, CAVLight [26], integrates cooperative perception and a cell transmission model to enhance observability under extremely low penetration (1%). Nevertheless, its reliance on calibrated flow parameters and lack of validation beyond a single intersection leaves open questions regarding network-level scalability.

Overall, while progress has been made in accommodating partial connectivity in DRL-based TSC, most existing studies remain limited to isolated intersections or rely on impractical simulation assumptions. Furthermore, relatively few works investigate network-level scalability while explicitly modeling realistic communication constraints.

III. METHODOLOGY

This section presents the proposed penetration-aware adaptive signal control framework. Figure 1 summarizes the overall workflow of the proposed framework. We begin by outlining the core max-pressure formulation and reinforcement learning structure underlying the controller. The approach is evaluated using a multi-agent simulation setup that couples the SUMO traffic simulator and LibSignal framework with OMNeT++ for communication modeling. The aim is to dynamically adjust policy parameters based on CV penetration levels in a mixed traffic environment.

A. Max-Pressure Formulation

Let $\mathcal{L}_{\text{in}}^i$ and $\mathcal{L}_{\text{out}}^i$ be the incoming and outgoing lane sets of intersection i , and $\mathcal{M}_i = \{(l, m) \mid l \in \mathcal{L}_{\text{in}}^i, m \in \mathcal{L}_{\text{out}}^i\}$ the admissible traffic movements. For any movement (l, m) , the *pressure* is the normalized density difference:

$$w_{l \rightarrow m}(t) = \frac{x_l(t)}{x_{\text{max},l}} - \frac{x_m(t)}{x_{\text{max},m}} \quad (1)$$

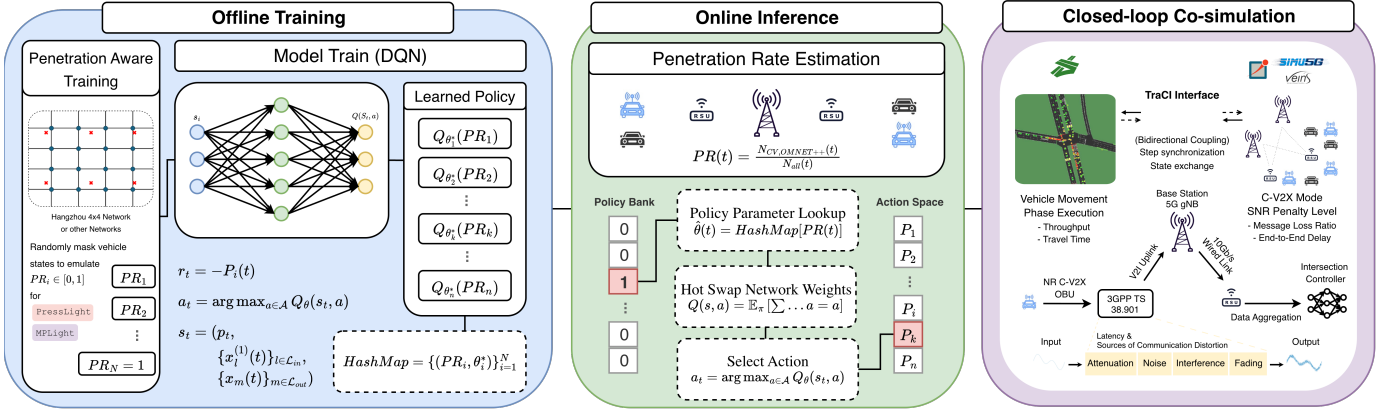


Fig. 1: Overview of the proposed penetration-aware traffic signal control framework. The real-time CV penetration rate $PR(t)$ dictates the dynamic selection of offline-optimized neural network weights $\hat{\theta}(t)$ for TSC.

where $x_l(t)$ is the queue length on lane l and $x_{\max,l}$ its jam capacity [27]. The intersection pressure is the sum of absolute movement pressures:

$$P_i(t) = \sum_{(l,m) \in \mathcal{M}_i} |w_{l \rightarrow m}(t)| \quad (2)$$

B. PressLight and MPLight

PressLight and MPLight are decentralized DRL-based models built on pressure-based traffic-state representations, providing a practical realization of *max-pressure control*. PressLight assigns an independent DQN agent to each intersection, using the max-pressure reward to reduce local queue imbalances. MPLight extends this idea with a shared-parameter multi-agent architecture based on the FRAP phase-competition representation, where intersections share a common Q-network during training while still selecting phases independently during execution. Thus, PressLight represents an independent-agent realization of pressure-based control, whereas MPLight provides a shared-network multi-intersection variant. Both models follow the same local decision rule:

$$a_i(t) = \arg \max_{a \in \mathcal{A}_i} Q_\theta^m(s_i(t), a) \quad (3)$$

where $m \in \{\text{PressLight}, \text{MPLight}\}$ denotes the evaluated model, $s_i(t)$ is the local traffic state at intersection i , $\mathcal{A}_i$ is the admissible phase set, and Q_θ^m is the corresponding model-specific Q-network.

a) *State*: The local state s_t for each intersection agent at time t is defined as a tuple:

$$s_t = (p_t, \{x_l^{(1)}(t)\}_{l \in \mathcal{L}_{in}}, \{x_m(t)\}_{m \in \mathcal{L}_{out}}) \quad (4)$$

where p_t is the currently active signal phase, $x_l^{(1)}(t)$ is the number of vehicles in the first segment of incoming lane l , and $x_m(t)$ is the number of vehicles on outgoing lane m . This formulation allows the agent to capture both queue accumulation and downstream congestion, essential for computing the pressure-based reward.

b) *Action*: The action a_t is a discrete selection of the next signal phase to apply at time t . PressLight supports a fixed phase plan (e.g., four-phase arterial), and the action space consists of all admissible signal phases:

$$a_t \in \mathcal{A} = \{\text{Phase}_1, \text{Phase}_2, \dots, \text{Phase}_N\} \quad (5)$$

c) *Reward*: The reward function aligns with the max-pressure formulation and is defined to penalize the total intersection pressure:

$$r_t = -P_i(t) \quad (6)$$

where $P_i(t)$ is the sum of absolute movement pressures at intersection i , as described in Eq. (2).

d) *Learning*: At each decision step, both PressLight and MPLight select the signal phase that maximizes the model-specific Q-value under the current local state:

$$a_t = \arg \max_{a \in \mathcal{A}} Q_\theta(s_t, a) \quad (7)$$

where $Q_\theta(s, a)$ is estimated by a Deep Q-Network. For PressLight, Q_θ^m is learned independently for each intersection, while for MPLight the Q-network parameters are shared across intersections using a phase-competition representation.

Each intersection is controlled by a DQN agent that estimates the optimal state-action value function:

$$Q_\theta(s, a) = \mathbb{E}_\pi \left[\sum_{\tau=0}^{\infty} \gamma^\tau r_{t+1+\tau} \mid s_t = s, a_t = a \right] \quad (8)$$

where s_t includes the current traffic signal phase, segmented queue lengths $\{x_l^{(1)}(t)\}$, and downstream queue lengths $\{x_m(t)\}$ and action a_t represents the selected signal phase to apply at time t . The function Q_θ is parameterized by a neural network with weights θ . It estimates the expected cumulative reward when action a is taken in state s . The network is trained by minimizing the Mean Squared Bellman Error (MSBE):

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s')} \left[\left(r + \gamma \max_{a'} Q_{\theta^-}(s', a') - Q_\theta(s, a) \right)^2 \right] \quad (9)$$

where θ^- denotes the parameters of a separate target network.

The target network Q_{θ^-} is updated less frequently than Q_{θ} to provide fixed targets for the MSBE calculation, thereby stabilizing training and preventing harmful feedback loops.

C. System Architecture and Deployment

From a system deployment perspective, the proposed framework employs centralized offline training and decentralized online execution. During the offline phase, DRL models are centrally trained for different CV PRs, and the optimized parameters are stored. During online deployment, each intersection is controlled by a local DRL agent that independently selects signal phases using traffic-state observations collected through its associated RSU. The RSUs serve exclusively as communication interfaces forwarding V2I data to the controller, while the cellular base stations support wireless communication without participating in DRL inference. Depending on the evaluated model, agents either share neural-network parameters during training (MPLight) or maintain distinct parameters (PressLight), while both preserve strictly decentralized execution at runtime.

D. Penetration-Rate-Aware Adaptation

During mixed-traffic operations, CV PR strongly influences the best-performing network weights θ^* . During the inference phase, we estimate the real-time PR as

$$PR(t) = \frac{N_{CV,OMNeT++}(t)}{N_{all}(t)} \quad (10)$$

where $N_{CV,OMNeT++}(t)$ denotes the number of CVs observed through V2I communication and $N_{all}(t)$ denotes the total number of vehicles detected by conventional traffic detectors deployed on the approaches of the controlled intersections. Such detectors may include inductive loop detectors, video-based detectors, or radar sensors that are commonly available in modern signalized intersections. The PR estimate therefore combines information from both the communication infrastructure and traditional sensing infrastructure. In this study, detector measurements are assumed to be available and error-free to isolate the effects of partial CV observability. In contrast, during training, we avoid the computational cost of simulating communication by randomly masking a portion of the vehicle state inputs to emulate different levels of observability. For each sampled penetration level PR_i , the evaluated DRL model (PressLight or MPLight) is trained until convergence, and the resulting mapping (PR_i, θ_i^*) is stored in a hash map. At runtime, the controller selects the appropriate neural network parameters via a lookup:

$$\hat{\theta}(t) = \text{HashMap}[PR(t)] \quad (11)$$

Since the models are trained for a discrete set of PRs, the controller selects the parameter set corresponding to the nearest available PR level stored in the policy bank. Following parameter selection, each agent hot-swaps its network weights $Q_{\hat{\theta}(t)} \leftarrow Q_{\theta(t)}$ before selecting $a_t = \arg \max_a Q_{\hat{\theta}(t)}(s_t, a)$. This adaptive parameter selection results in penetration-aware

Algorithm 1 : PR-Aware Adaptive Policy Control Loop

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1: Offline phase:
2: for each penetration rate  $PR_i$  do
3:   Train the model and store optimal weights  $\theta_i^*$ 
4: end for
5: Store  $(PR_i, \theta_i^*)$  pairs in a hash map for runtime lookup
6: Online control loop:
7: for each control timestep  $t$  do
8:   Estimate penetration rate  $PR(t)$ 
9:   Retrieve adapted weights  $\hat{\theta}(t) = f(PR(t))$ 
10:  Set  $Q \leftarrow Q_{\hat{\theta}(t)}$ 
11:  Select action  $a_t = \arg \max_a Q(s_t, a)$ 
12:  Apply  $a_t$  and observe  $r_t$  and  $s_{t+1}$ 
13: end for

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variants of the evaluated DRL-based TSC models, denoted as *PR-PressLight* and *PR-MPLight*.

Algorithm 1 outlines the adaptive control loop. Offline, the model is trained for discrete penetration levels PR_i , storing optimal parameters θ_i^* in a hash map. Online, the current penetration rate $PR(t)$ is estimated to retrieve the adapted weights $\hat{\theta}(t) = f(PR(t))$. The Q-network is dynamically updated with these weights to select the optimal action $a_t = \arg \max_a Q(s_t, a)$, and the process repeats for subsequent timesteps.

E. Communication Network Simulation Model

The communication flows from vehicles to intersections are simulated using a federated telco-traffic simulation approach [8]. Individual vehicles in SUMO traffic simulator [9] are synchronized with a communication simulation platform entitled Objective Modular Network Testbed in C++ (OMNeT++) [7] via SUMO's TraCI bidirectional coupling interface [9]. A modified version of Veins simulation framework [8] is used to provide TraCI Application Programming Interface implementation for OMNeT++. The message transmissions from vehicles are assumed to be carried out by 5G-based C-V2X communication [28]. To simulate the V2I message transmissions, we use Simu5G [10] and INET [7] model libraries for OMNeT++. The transmissions are carried out in NR C-V2X infrastructural mode, where vehicles send messages to the RSU through the 5G gNodeB base station, which also allocates communication resources to communicating devices to avoid conflicts in resource scheduling. To cover the investigated area of 16 intersections with total dimensions of 4×3 kilometres, 6 cellular base stations were used in a grid placement as indicated in Figure 2. The number of serving base stations has been selected in accordance with earlier investigations conducted in [29]. Each base station is a macro cell equipped with an omnidirectional antenna and transmits with a power of 46 dBm. Vehicles are equipped with an omnidirectional antenna and transmit with 26 dBm of power, over a channel with 5MHz bandwidth. The message length is set to 300 Bytes and the message generation period is 10 messages per second as outlined by the ETSI EN 302 637-2 [30] specification

of Cooperative Awareness Basic Service. Each intersection is equipped with a C-V2X capable Roadside Unit (RSU), which receives vehicle messages and provides information about known vehicles to intersection controller via a lossless link. A placement of the RSUs is depicted in Figure 2.

To model the impact of channel-quality degradation caused by attenuation, interference, noise, and fading, we use the Signal-to-Noise Ratio (SNR) Penalty parameter proposed in [4]. We vary the SNR Penalty value from 0 to 25 dB to evaluate the impact of deteriorating channel conditions on traffic-control performance. The penetration of CVs in the simulation is controlled by the PR parameter in Veins according to the uniform distribution in range of 0-1. The successful message transmissions are logged in a shared file together with basic information about the message, such as timestamps of message transmission and reception, transmission end-to-end delay and vehicle position at the time of message transmission.

In the proposed framework, traffic-state observability is directly coupled with the quality and availability of V2I communication. The controller constructs its state representation using information received from CVs through the RSUs. Consequently, lower PRs, higher packet losses and communication delays, or degraded wireless conditions reduce the completeness and freshness of the observed traffic state. These limitations create partial observability for the DRL controller, because the agent is supposed to select signal phases based on incomplete or delayed information rather than a full traffic state.

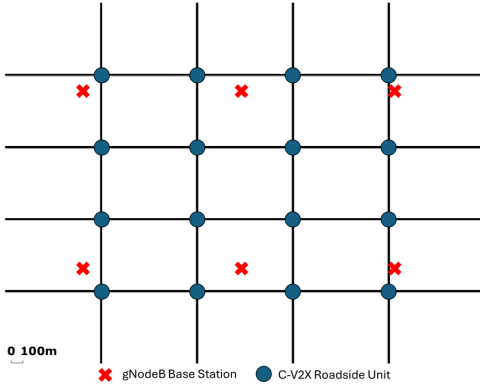


Fig. 2: Layout of the communication network.

IV. RESULTS

To evaluate the proposed framework, experiments were conducted over a 1-hour simulation horizon using a 1-s simulation step and a 30-s signal control interval. DRL agents were trained for 100 episodes on the Hangzhou 4×4 network. For each PR configuration, five independent runs with different random seeds were conducted, and reported metrics show the mean and 95% confidence interval. The throughput sensitivities under varying SNR Penalty and packet-level statistics for different PRs were collected over 900-s horizon for message loss and end-to-end message delay analysis, by again five

independent runs (different random seeds) and reporting the mean and 95% confidence interval.

Figure 3 shows that partial observability severely degrades the baseline controllers, making the low penetration range (0% to 20%) the most critical deployment regime. At 5% penetration, PR-PressLight provides the largest benefit, improving throughput by 9.8% (from 2230 to 2448 vehicles) and reducing average travel time by 45.5s compared to baseline PressLight. Averaged up to 20% penetration, PR-PressLight reduces travel time by 11.1% and improves throughput by 6.8%. While MPLight exhibits stronger baseline robustness, the PR-aware variant still yields benefits, reducing travel time by 23.4s at 5% penetration and smoothing performance volatility. Across the low penetration range, PR-MPLight reduces average travel time by 4.5%. As penetration increases, the gap between baseline and PR-aware models narrows, converging entirely at full penetration. These results confirm that matching the policy to the observed penetration level mitigates performance volatility without degrading performance when complete CV information is available.

To evaluate robustness under communication degradation, we compared models trained with PR values of 0.1, 0.5, and 1.0 across various SNR penalty settings (Figs. 4 and 5). Throughput generally decreases as the SNR penalty increases due to poorer V2I link quality. Under mild degradation, the $PR = 1.0$ model performs best. However, as SNR penalties rise and packet losses lower the effective penetration, the effective observability approaches that experienced under lower PR conditions (e.g., $PR = 0.5$ and $PR = 0.1$), allowing these corresponding models to become more robust and outperform $PR = 1.0$. This demonstrates that robustness under degraded V2I links depends heavily on matching the training PR to the effective deployment penetration.

V. DISCUSSION AND CONCLUSION

The experiments demonstrate that partial vehicle connectivity significantly reshapes the effectiveness and stability of DRL-based TSC. In low penetration conditions, baseline models experience severe performance degradation and volatility. In contrast, our penetration-aware adaptation mechanism remains resilient by dynamically supplying each agent with offline-computed parameters pre-specialized for the current PR. This confirms that agencies can capture the benefits of DRL even before CV technology becomes widespread.

The proposed framework delivers two practical contributions. First, the open-source co-simulation platform allows researchers to evaluate learning-based controllers under realistic wireless conditions. Second, the adaptation module improves robustness across different connectivity levels while preserving decentralized execution.

For practical deployment, two timing factors are most relevant: V2I communication latency and DRL inference latency. Communication performance was evaluated through the OM-NeT++/Simu5G packet-level results, while controller runtime was profiled separately in SUMO-only inference mode. The DRL inference profiled runs show that both controllers operate

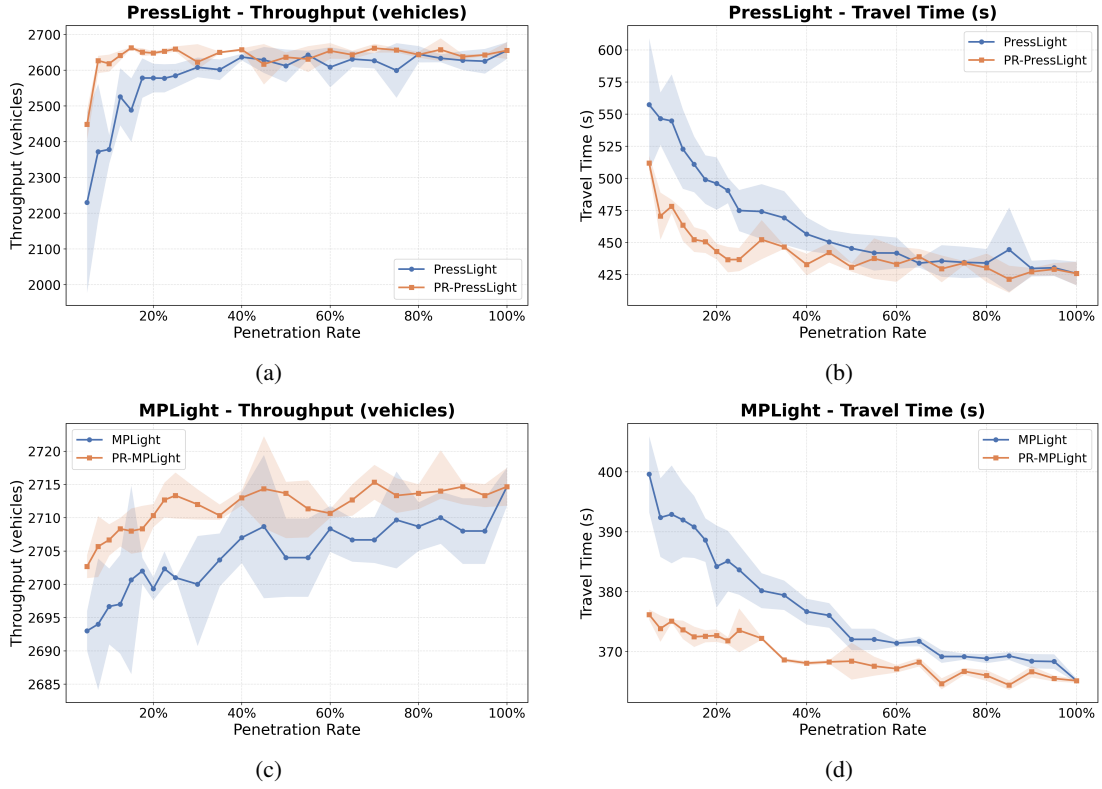


Fig. 3: Performance comparison across connected-vehicle penetration rates: (a) PressLight throughput, (b) PressLight travel time, (c) MPLight throughput, and (d) MPLight travel time.

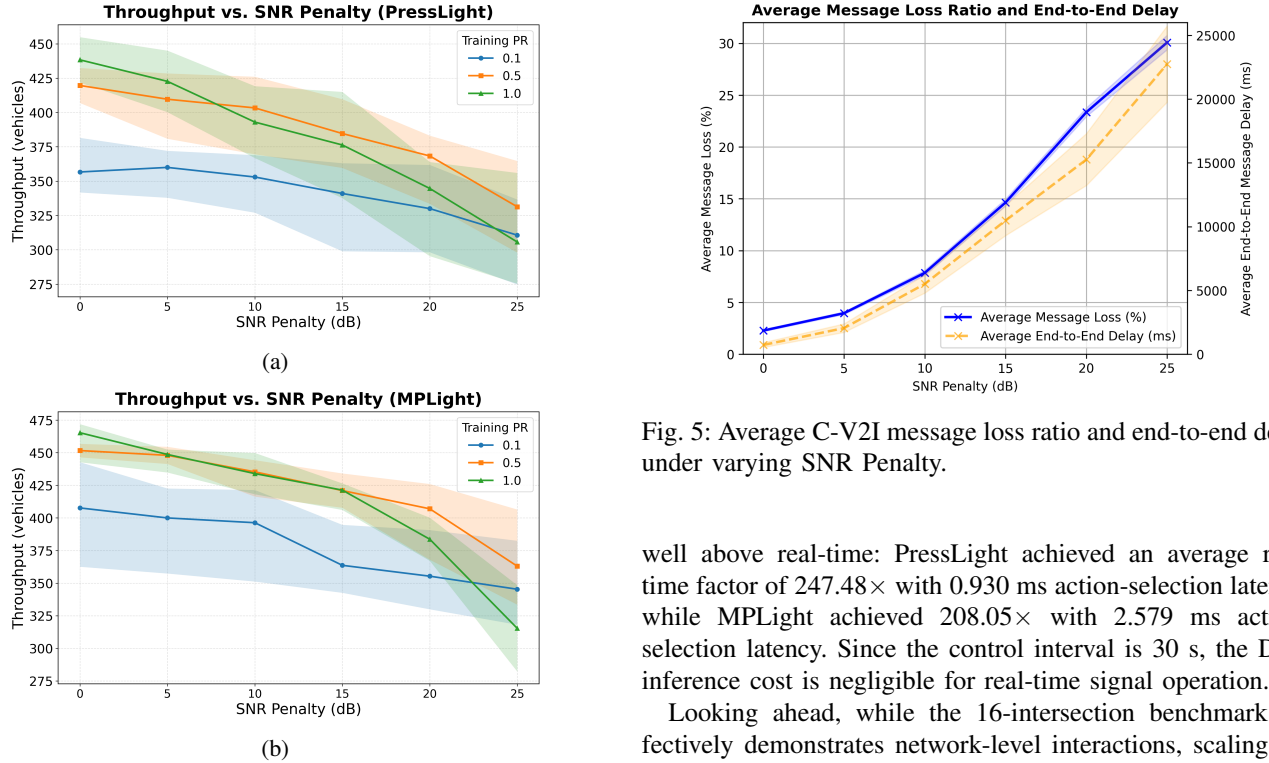


Fig. 4: Throughput sensitivity under varying SNR Penalty: (a) PressLight and (b) MPLight.

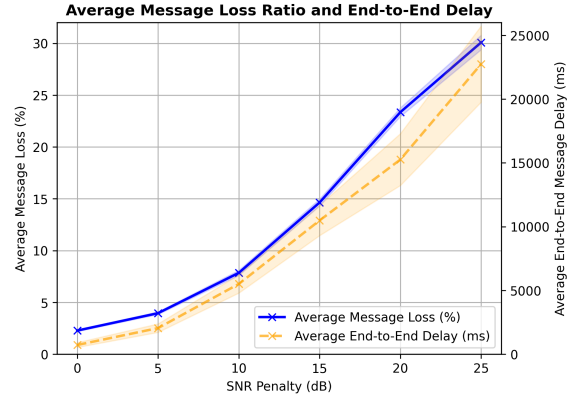


Fig. 5: Average C-V2I message loss ratio and end-to-end delay under varying SNR Penalty.

well above real-time: PressLight achieved an average real-time factor of $247.48\times$ with 0.930 ms action-selection latency, while MPLight achieved $208.05\times$ with 2.579 ms action-selection latency. Since the control interval is 30 s, the DRL inference cost is negligible for real-time signal operation.

Looking ahead, while the 16-intersection benchmark effectively demonstrates network-level interactions, scaling the framework to larger and more heterogeneous urban topologies remains for future work. Such extensions will require redesigning the communication layout, retraining the DRL models,

and evaluating the computational overhead of larger-scale SUMO-OMNeT++ co-simulation. In addition, the current deployment analysis profiles DRL inference latency separately from the live packet-level co-simulation. A complete end-to-end runtime study, including OMNeT++ execution overhead, RSU message processing, and controller inference in one synchronized loop, will be added in future work.

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REFERENCES

- [1] W. Sleiman, P. Beigi, and S. Hamdar, “Percolation-Based Cloud Computing Framework for Network Reliability Assessment,” in *2025 IEEE 4th International Conference on Computing and Machine Intelligence (ICMI)*. IEEE, 2025, pp. 1–7.
- [2] H. Wei, C. Chen, G. Zheng, K. Wu, V. Gayah, K. Xu, and Z. Li, “Presslight: Learning max pressure control to coordinate traffic signals in arterial network,” in *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*, 2019, pp. 1290–1298.
- [3] H. Wei, C. Chen, X. Xu, and et al., “Colight: Learning network-level cooperation for traffic signal control,” *Proceedings of the 28th ACM International Conference on Information and Knowledge Management*, 2019.
- [4] I. Finkelberg, T. Petrov, A. Gal-Tzur, N. Zarkhin, P. Počta, T. Kováčiková, L. Buzna, M. Dado, and T. Toledo, “The effects of vehicle-to-infrastructure communication reliability on performance of signalized intersection traffic control,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 9, pp. 15 450–15 461, 2022.
- [5] W. Sleiman, P. Beigi, T. Petrov, L. Buzna, P. Počta, and S. Hamdar, “Impact of Pedestrian and Vehicle Connectivity on Intersection Performance: A High-Fidelity Simulation with Deep Reinforcement Learning Control,” *Transportation Research Record*, pp. 1–17, 2025.
- [6] H. Mei, X. Lei, L. Da, B. Shi, and H. Wei, “Libsignal: An open library for traffic signal control,” *Machine Learning*, vol. 113, no. 8, pp. 5235–5271, 2024.
- [7] A. Varga, “OMNeT++,” in *Modeling and tools for network simulation*. Springer, 2010, pp. 35–59.
- [8] C. Sommer, R. German, and F. Dressler, “Bidirectionally Coupled Network and Road Traffic Simulation for Improved IVC Analysis,” *IEEE Transactions on Mobile Computing*, vol. 10, no. 1, pp. 3–15, January 2011.
- [9] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. Lücken, J. Rummel, P. Wagner, and E. Wiessner, “Microscopic traffic simulation using sumo,” in *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*, 2018, pp. 2575–2582.
- [10] G. Nardini, D. Sabella, G. Stea, P. Thakkar, and A. Virdis, “Simu5G—An OMNeT++ Library for End-to-End Performance Evaluation of 5G Networks,” *IEEE Access*, vol. 8, pp. 181 176–181 191, 2020.
- [11] J. V. S. Busch, V. Latzko, M. Reisslein, and F. H. P. Fitzek, “Optimised Traffic Light Management Through Reinforcement Learning: Traffic State Agnostic Agent vs. Holistic Agent With Current V2I Traffic State Knowledge,” *IEEE Open Journal of Intelligent Transportation Systems*, vol. 1, pp. 201–216, 2020.
- [12] H. Wei, G. Zheng, H. Yao, and Z. Li, “IntelliLight: A Reinforcement Learning Approach for Intelligent Traffic Light Control,” *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery Data Mining*, p. 2496–2505, 2018.
- [13] G. Zheng, H. Wei, C. Chen, and et al., “MPLight: Learning an Efficient and Coordinated Traffic Signal Control Policy,” *Proceedings of the 28th International Joint Conference on Artificial Intelligence (IJCAI)*, 2020.
- [14] X. Xu, H. Wei, C. Chen, G. Zheng, and et al., “DACT: A Deep Reinforcement Learning Framework for Adaptive and Cooperative Traffic Signal Control,” *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*, 2021.
- [15] M. Vieira, M. A. Vieira, G. Galvão, P. Louro, M. Véstias, and P. Vieira, “Enhancing Urban Intersection Efficiency: Utilizing Visible Light Communication and Learning-Driven Control for Improved Traffic Signal Performance,” *Vehicles*, vol. 6, no. 2, pp. 666–692, 2024.
- [16] W. Sleiman, M. Haque, M. S. Amin, P. Beigi, M. Khoueiry, S. Kang, and S. Hamdar, “Diffusion Process-Based Model for Network Trajectory Propagation: Formulation, Data Mining, and Cross-Comparative Analysis,” *IEEE Transactions on Intelligent Transportation Systems*, 2026.
- [17] R. Zhang, A. Ishikawa, W. Wang, B. Striner, and O. K. Tonguz, “Using Reinforcement Learning With Partial Vehicle Detection for Intelligent Traffic Signal Control,” *IEEE Trans. Intelligent Transportation Systems*, vol. 22, no. 1, pp. 404–415, 2021.
- [18] Y. Feng, J. Zheng, and H. X. Liu, “Real-Time Detector-Free Adaptive Signal Control with Low Penetration of Connected Vehicles,” *Transportation Research Record*, vol. 2672, no. 18, pp. 35–44, 2018.
- [19] Y. Wang, M. Wang, W. Daamen, and S. Hoogendoorn, “Development and Evaluation of an Adaptive Traffic Signal Control Scheme Under a Mixed-Automated Traffic Scenario,” in *2021 IEEE International Intelligent Transportation Systems Conference (ITSC)*. IEEE, 2021, pp. 2484–2489.
- [20] R. Ducrocq and N. Farhi, “Using Reinforcement Learning With Partial Vehicle Detection for Intelligent Traffic Signal Control,” *arXiv preprint arXiv:2201.03962*, 2022.
- [21] L. Song and W. Fan, “Traffic Signal Control Under Mixed Traffic With Connected and Automated Vehicles: A Transfer-Based Deep Reinforcement Learning Approach,” *IEEE Access*, vol. 9, pp. 145 228–145 237, 2021.
- [22] G. Zhang, H. Huang, and F. Chang, “TS-PVL: Two-Stage Deep-Reinforcement-Learning-Based Traffic Light With Pedestrian-Vehicle Control in Mixed-Autonomy Traffic,” *IEEE Internet of Things Journal*, vol. 12, no. 15, pp. 31 001–31 015, 2025.
- [23] S. M. A. B. A. Islam, A. Hajbabaie, and H. M. A. Aziz, “A real-time network-level traffic signal control methodology with partial connected vehicle information,” *Transportation Research Part C: Emerging Technologies*, vol. 121, p. 102830, 2020.
- [24] M. Zhao, W. Li, Y. Fu, K. Ruan, and X. Di, “CVLight: Decentralized learning for adaptive traffic signal control with connected vehicles,” *Transportation Research Part C: Emerging Technologies*, vol. 141, p. 103728, 2022.
- [25] J. Guo, L. Cheng, and S. Wang, “CoTV: Cooperative Control for Traffic Light Signals and Connected Autonomous Vehicles using Deep Reinforcement Learning,” *arXiv preprint arXiv:2201.13143*, 2023.
- [26] W. Li, T. Zhu, and Y. Feng, “A cooperative perception based adaptive signal control under early deployment of connected and automated vehicles,” *Transportation Research Part C: Emerging Technologies*, vol. 169, p. 104860, 2024.
- [27] P. Varaiya, “Max pressure control of a network of signalized intersections,” *Transportation Research Part C: Emerging Technologies*, vol. 36, pp. 177–195, 2013.
- [28] T. Petrov, W. Sleiman, P. Beigi, P. Počta, L. Buzna, and S. Hamdar, “Simulation of Vehicle-to-Pedestrian Communication for Development of Novel C-ITS Services for Vulnerable Road Users,” *Transportation Research Procedia*, vol. 93, pp. 674–679, 2026.
- [29] T. Petrov, P. Počta, and T. Kovacikova, “Benchmarking 4G and 5G-Based Cellular-V2X for Vehicle-to-Infrastructure Communication and Urban Scenarios in Cooperative Intelligent Transportation Systems,” *Applied Sciences*, vol. 12, no. 19, 2022.
- [30] European Telecommunications Standards Institute, “ETSI EN 302 637-2: Intelligent Transport Systems (ITS); Vehicular Communications; Basic Set of Applications; Part 2: Specification of Cooperative Awareness Basic Service,” European Telecommunications Standards Institute, Sophia Antipolis Cedex, France, Tech. Rep., Sep. 2014.